INTRODUCTION TO MATHEMATICAL ANALYSIS MATH2241
MIDTERM TEST

Answer all four questions. Full score is 40pts.

(1) (10pts) Let (a_n) be a convergent sequence. Let (x_n) be a sequence of positive numbers such that $x_n^2 - a_n x_n - 1 = 0$ for all $n \in \mathbb{N}$. Prove that (x_n) is convergent.

(2) (10pts) Prove that every sequence (a_n) has a monotone subsequence.

(3) (10pts) Prove or disprove the convergence of the series $(s_n = \sum_{k=1}^{n} \frac{k!}{k^k})$.

(4) (10pts) Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $\lim_{x\to 0} f(x) = L \in \mathbb{R}$. Prove that

$$\lim_{x\to 0} f(x^2) = L.$$